

1. Testbench Result

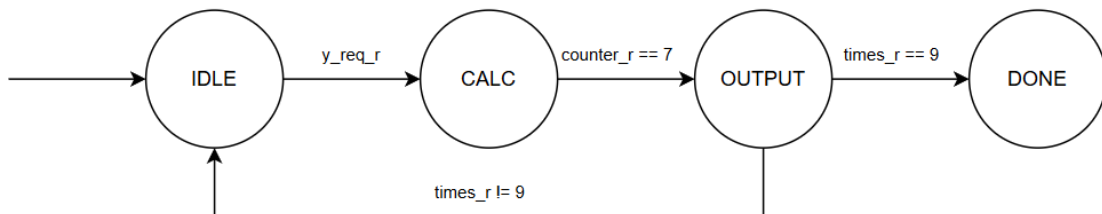
```
-----The test result is .....Correct -----
```

Pass!!!!!!!!!!!!!!

2. Check Design Result

Register Name	Type	Width	Bus	MB	AR	AS	SR	SS	ST
state_r_reg	Flip-flop	2	Y	N	Y	N	N	N	N
id_dist_reg	Flip-flop	24	Y	N	Y	N	N	N	N
d_dist_reg	Flip-flop	112	Y	N	Y	N	N	N	N
OutData_r_reg	Flip-flop	24	Y	N	Y	N	N	N	N
out_valid_r_reg	Flip-flop	1	N	N	Y	N	N	N	N
y_req_r_reg	Flip-flop	1	N	N	Y	N	N	N	N
times_r_reg	Flip-flop	4	Y	N	Y	N	N	N	N
counter_r_reg	Flip-flop	3	Y	N	Y	N	N	N	N

3. Implementation (I'm used to using “_r” for the now state while “_w” for the next state)



State	Explanation
IDLE	Wait for y_req to rise to start calc
CALC	Calculate the pred_y, distance, and sort the distance
OUTPUT	Output the answer
DONE	Done

Variable	Explanation
y_req	Being 1 when needs to accept the data
counter	Count the number of the data
times	Count the number of the testing data

(1) Calculate the data:

For the eight x's, I calculated them separately, which makes the variable less but takes more time to calculate. Hence, there is a variable called "counter" to help me count which x is calculated now. When counter == 7, which means all eight x's are calculated, then we can output the result.

(2) Sorting

Since there is not much data, I use bubble sort to sort the data.

(3) Times

There are 10 testing data in total, so there is a variable named "times" to help me memorize how many testing data I have been working with.